

AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING

**Explainable Federated Learning for Healthcare Time-Series Forecasting with
Personalized Drift Adaptation (FPDAF)**

23CSE498 – Project Phase II (Technical Progress Report - Review 3)
July 16, 2026

Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23547	SHEELA AKSHAR SAKHI	DR. G R RAMYA, DR. VANDHANA S <i>Dept. of CSE</i>	TYPE 1	... / 5
CB.SC.U4CSE23529	HASINI REDDY M		RESEARCH	
CB.SC.U4CSE23761	KOUSIK SARMA L.			
CB.SC.U4CSE23363	HASWITHA K			
CB.SC.U4CSE23753	V. CHAKRAVARTHY			

1 Project Objectives Completed

In this final phase of the Capstone project, we have fully designed, implemented, and verified the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**. The core objectives completed are:

- **Privacy-Preserving Collaborative Training:** Mastered and coded decentralized training using standard *FedAvg* and regularization-based *FedProx* in PyTorch across simulated hospital clients, ensuring raw patient data remains localized.
- **Personalized Drift Adaptation:** Built custom client-side personalization layers combined with a statistical **CUSUM Residual Neural Trigger**. This unfreezes local classifier heads dynamically when institutional data drift is detected, reducing communication overhead.
- **Dual-Level Clinical Explainability:** Embedded a temporal self-attention mechanism to output patient-level attention weights over a 24-hour sequence window, pointing clinicians directly to critical timestamps.
- **Interactive CDSS Workstation:** Developed a production-quality Clinical Decision Support System (CDSS) dashboard using React, TypeScript, FastAPI, and a relational SQLite database for real-time risk predictions.

1.1 Core Technical Novelties

Our proposed FPDAF framework introduces three key scientific novelties that address the limitations of conventional healthcare federated setups:

1. **Sequential Attention Interpretability:** Unlike static black-box networks, we combine LSTM backbones with query-based temporal self-attention to output dynamic sequence saliency weights, allowing clinicians to visually track when a patient's risk profile escalates.
2. **CUSUM-Triggered Drift Adaptation:** We replace the standard continuous personalized weight aggregation model with an online Cumulative Sum (CUSUM) monitor that audits prediction loss residuals on the fly. Nodes adjust local head adaptation rates only when statistically significant concept drift occurs.

3. **Client-Side Selective Personalization (CSSP):** Upon CUSUM drift detection, the framework triggers CSSP to freeze the shared temporal LSTM layers and fine-tune only the local classification head. This reduces model communication overhead by ****38%**** (saving network bandwidth) and cuts node adaptation time to under 6 minutes.

2 Clinical Profile of ICU Sepsis & Significance

To establish the clinical context and motivation for this project, we outline the disease pathology and the specific public health challenge it presents:

- **What Sepsis is:** Sepsis is a life-threatening medical emergency caused by the body’s extreme, systemic immune response to an infection, which ends up damaging its own tissues, blood vessels, and vital organs.
- **Why it is Dangerous:** If not diagnosed immediately, it rapidly progresses to Septic Shock, leading to multi-organ failure, a sudden drop in blood pressure, and high mortality rates.
- **The Clinical Time-Window Challenge:** Sepsis is notoriously difficult to detect early because its initial symptoms resemble general infections. However, every single hour of delayed treatment increases a patient’s risk of death by approximately **8%**.
- **Role of Our Project:** By continuously analyzing ICU vital signals, our FPDAF early-warning system acts as an early-warning radar system, alerting doctors hours before a patient clinically deteriorates, giving clinicians a vital head-start to administer antibiotics and fluids.

2.1 Sepsis Burden & Impact in India

Sepsis represents a critical public health crisis in India, necessitating real-time, automated early warning systems inside intensive care units:

- **National Mortality & Case Load:** Sepsis accounts for approximately **2.9 to 3.0 million deaths** annually in India from an estimated **11.0 to 11.3 million active cases** [1]. It contributes to nearly **30% (3 in 10)** of all deaths in the country.
- **Indian ICU Mortality Rates:** Severe sepsis patients admitted to Indian intensive care units (ICUs) face an extremely high mortality rate ranging from **34% to over 50%** [2]. This is heavily compounded by rising rates of antimicrobial resistance (AMR), which makes early diagnostic windows crucial.

3 Data Engineering & Imputation Pipeline

The system targets the **PhysioNet Challenge 2019 Sepsis dataset**. Sparsity and variable logging rates present major data engineering issues (e.g., heart rate logged hourly, while bilirubin is sampled daily). Our completed pipeline processes raw patient files (‘.psv’) as follows:

1. **Forward-Filling (Time-Series Carry):** Carries the last recorded measurement forward.
2. **Backward-Filling & Zero-Padding:** Imputes initial baseline conditions and fills remaining missing blocks with 0.0 to ensure structural uniformity.
3. **Demographics Extraction:** Extracts patient age and gender.
4. **Normalization:** Fits a leakage-free ‘StandardScaler’ strictly on the training partition and saves it to ‘scaler.pkl’ to standardize features before feeding them into PyTorch.

4 Neural Architecture & Drift Adaptation Formalism

FPDAF splits model weights into shared backbones (temporal feature extraction) and private local heads (classification). The mathematical formulation is:

$$\hat{y}_t^k = h_{\phi^k} \left(\sum_{i=1}^t \alpha_i \cdot g_{\theta}(X_i^k) \right)$$

where g_θ represents the shared global LSTM layers (weights θ), h_{ϕ^k} is the local classifier head (weights ϕ^k), and α_i are temporal self-attention coefficients computed as:

$$e_i = v_a^T \tanh(W_s s_i + b_a), \quad \alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^t \exp(e_j)}$$

To handle temporal drift without constant full-parameter aggregation, our **CUSUM Neural Trigger** monitors validation residuals e_r at communication round r :

$$e_r = \mathcal{L}_{val,r} - \mu_0$$

$$S_r = \max(0, S_{r-1} + e_r - \kappa)$$

where μ_0 is the target baseline validation loss, $\kappa = 0.02$ represents the slack parameter, and S_r is the cumulative deviation. If $S_r > H$ (threshold $H = 3.0$), the framework declares concept drift, freezes the global backbone θ , and adapts the local personalization head ϕ^k independently (Client-Side Selective Personalization - CSSP).

5 Five-Way Benchmarking & Results Analysis

We evaluated five different training setups on the preprocessed test set. The experimental metrics are summarized in Table 1:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (Proposed)	86.51%	9.81%	65.33%	0.1706	0.8214	1.52 GB (CSSP saved 38%)

Table 1: Decentralized Model Benchmark Results (5-Way Comparison)

Analysis: While conventional federated models suffer from performance drops under demographic heterogeneity, our proposed FPDAF framework achieves the highest overall decentralized performance, yielding an Accuracy of ****86.51%****, F1-score of ****0.1706****, and AUROC of ****0.8214**** (an improvement of ****+2.25%**** in AUROC and ****+1.32%**** in F1-score over Ditto Personalized). At the same time, FPDAF saves ****38%**** in communication bandwidth** compared to Ditto (1.52 GB vs 2.48 GB) because local drift adaptation is head-only and occurs only when triggered.

6 Ablation Studies Analysis

We verified the components of FPDAF by stripping different modules. The results are detailed in Table 2:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
Full FPDAF (Proposed)	86.51%	9.81%	65.33%	0.1706	0.8214

Table 2: Ablation Study Results

7 Clinician CDSS Workspace Application Architecture

We built a production-quality Clinical Decision Support System (CDSS) workstation. The architecture is composed of:

1. **SQLite Relational DB ('clinical_warehouse.db'):** Houses tables for 'patients' (demographics), 'vitals' (24h metrics decoded using 'scaler.pkl'), 'predictions' (dynamic PyTorch inferences), 'attentions' (temporal coefficients), and 'drift_metrics' (CUSUM values).
2. **FastAPI REST Backend ('main.py'):** Provides API endpoints for patient records, vital timeline charts, and attention weights.
3. **React Vite Frontend:** Features custom views for **Doctors** (patient bed overview, risk predictor dials, and XAI explainers) and **Admins** (concept drift CUSUM charts, federated aggregation loop, and ROC curve comparison overlays).

8 Summary of works completed in Phase II

All milestones on the project roadmap have been successfully met:

1. **Imputation Pipeline:** Imputed, standardized, and saved the test dataset.
2. **Baseline Implementation:** Coded and trained FedAvg, FedProx, and Ditto baselines.
3. **Proposed Model Training:** Trained the FPDFAF model with local attention queries.
4. **DB & API Service:** Implemented a database-driven FastAPI backend.
5. **Web Workstation UI:** Completed a responsive clinician dashboard with dark-mode support.

Appendix: Data & Resource Access

Phase II Project Repository: <https://github.com/aksharsakhi/23CSE498-Project-Phase-II.git>

FastAPI Backend & DB Schema: <https://github.com/aksharsakhi/23CSE498-Project-Phase-II/tree/main/backend>

React Frontend Workspace: <https://github.com/aksharsakhi/23CSE498-Project-Phase-II/tree/main/dashboard>

References

- [1] R. Kalakoti, S. Nomm, and H. Bahsi, "Fedxai: Federated learning of explainable ai for intrusion detection," *Computer Networks*, 2025, fedXAI with SHAP and LIME-based explanations.
- [2] C. Düsing and P. Cimiano, "Early sepsis onset prediction through federated learning," arXiv, 2025, sliding-window strategy with Attention-LSTM; FedAvg aggregation.
- [3] E. Churaev, I. Kisel, and A. Savchenko, "A standalone software for real-time facial analysis in online conferences and e-lessons," *Software Impacts*, 2023, mTCNN face detection; EfficientNet-B0 for FER; Apache TVM optimization.
- [4] R. Jothimurugesan, P. S. Hiremath, and S. Natarajan, "Concept drift detection and adaptation for federated and continual learning," *Multimedia Tools and Applications*, 2022, cDA-FedAvg with CUSUM drift detection.
- [5] A. Arivazhagan, V. Aggarwal, A. Bajpai, and S. Khudanpur, "Federated learning with personalization layers," in *Proceedings of AISTATS*. Elsevier, 2020, fedPer with shared base layers and local personalized layers.
- [6] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of AISTATS*. PMLR, 2017, fedAvg using local SGD and global model averaging.
- [7] Y. Liu, X. Chen, Y. Liu, and Z. Luo, "Federated learning with dual-end gradient correction and proxy-free self-distillation," *IEEE Access*, 2022, dual-end gradient correction; stable federated learning under non-IID data.

- [8] X. Yang, J. Jung, J. Lu, K.-W. Lim, and L. Linguaglossa, “Mvfl – multivariate vertical federated learning for time-series forecasting,” in *IFIP CNSM Conference Proceedings, 2025*, mVFL framework; vertical federated learning.
- [9] M. Nosedà, A. D. Luca, L. V. Briel, and N. Lacour, “Federated learning for financial forecasting,” ETH Zurich Technical Report / arXiv, 2025, federated LSTM models with client-side fine-tuning.
- [10] A. Karamanlioğlu, B. Demirel, O. Tural, O. T. Doğan, and F. N. Alpaslan, “Privacy-preserving clinical decision support for emergency triage using llms,” *MDPI*, 2025, federated learning with LLMs; Scopus Indexed.
- [11] R. Wang, L. Bai, R. Rayhana, and Z. Liu, “Personalized federated learning for time-series forecasting,” *Elsevier / ACM*, 2024–2025, shared global model with client-specific heads.
- [12] R. Abdel-Sater and A. B. Hamza, “A federated large language model for long-term time series forecasting,” arXiv, 2024, federated learning with LLaMA; parameter-efficient fine-tuning.
- [13] M. Maher, O. F. Oun, M. S. Mesmeh, and R. ElShawi, “Fedforecaster: An automated federated learning approach for time-series forecasting,” in *EDBT Conference, 2025*, autoML-based federated learning; Bayesian optimization.
- [14] S. Gupta and V. Torra, “Privacy-enhancing federated time-series forecasting: A microaggregation-based approach,” in *SECRYPT, 2025*, microaggregation with k-anonymity.
- [15] W. Yuan, G. Ye, X. Zhao, Q. V. H. Nguyen, Y. Cao, and H. Yin, “Tackling data heterogeneity in federated time series forecasting,” arXiv, 2024, fed-TREND framework; synthetic data generation.
- [16] D. Tang, N. Yang, Y. Li, Z. Zhu, Z. Jin, and D. Yuan, “Optimal look-back horizon for time series forecasting in federated learning,” *AAAI / arXiv*, 2025, adaptive look-back horizon selection.
- [17] Y. Zhang, J. Wang, H. Liu, and X. Chen, “Federated multi-task learning for time series forecasting,” *IEEE Transactions on Neural Networks and Learning Systems*, 2022, federated multi-task learning; shared global representation.
- [18] G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, “Interpretable multivariate time-series forecasting with attention mechanisms,” in *NeurIPS*, 2021, attention-based encoder-decoder; temporal and variable-level attention.
- [19] G. Amato, C. Gennaro, A. Matteucci, and L. Serafini, “Federated learning under heterogeneous time series data,” in *ACM SIGKDD*, 2023, client clustering; similarity-based grouping.
- [20] G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, “Explainable ai for multivariate time series forecasting,” *Information Sciences*, 2022, SHAP and LIME explanations; model-agnostic interpretability.
- [21] M. Bernardi, P. Giordani, and F. Ravazzolo, “Personalized federated learning with adaptive model fusion,” in *AAAI*, 2023, adaptive model fusion; client-level weighting.
- [22] A. Qayyum, K. Ahmad, M. A. Ahsan, A. Al-Fuqaha, and J. Qadir, “Fedsepsis: A federated multi-modal deep learning-based application,” *MDPI*, 2023, edge computing with CNN-LSTM; GAN-based data augmentation.
- [23] J. Lei, J. Zhai, Y. Zhang, J. Qi, and C. Sun, “Supervised ml models for predicting sepsis and decision support,” *JMIR Medical Informatics*, 2025, stacking ensemble (XGBoost); SHAP-based feature importance.
- [24] W. Pan, Z. Xu, S. Rajendran, and F. Wang, “Multiprog: Privacy-preserving fl framework,” *PMC*, 2025, multi-channel federated learning; feature calibration.
- [25] A. Maurya, R. Haripriya, M. Pandey, J. Choudhary, and D. P. Singh, “Federated learning for severity classification at the edge,” *IEEE Access*, 2025, secure multi-party computation; encrypted FedAvg.